

SeismoFNO: Physics-Grounded Neural Operators for Seismic Structural Response Prediction

A comprehensive walkthrough of motivation, methods, failure diagnostics, and open problems

Raghvendra Singh Gahlot · B.Tech (Civil Engineering, Structural Engineering) · September 2026

Why I started this

This project began after running a series of nonlinear time-history analyses in OpenSeesPy for a seismic design course and observing how long a single simulation took to converge—even for a relatively small five-story shear frame. Regional risk assessment, city-scale loss estimation, and performance-based design optimization require evaluating these response histories across thousands of distinct structures and ground motions. The computational gap between rigorous numerical simulation and practical engineering applications motivated exploring whether a learned neural operator surrogate could bridge the divide.

What emerged along the way was a compelling technical narrative: the standard Scientific ML recipe—Fourier Neural Operators on a Cartesian lattice—breaks down when applied to structures with variable story counts. Diagnosing that failure, developing a topology-native graph operator, and conditioning it on pre-earthquake modal physics form the core of this work. Throughout this investigation, our goal has been scientific honesty: documenting what breaks is just as critical as documenting what works.

1. Foundations: rigorous ground-truth validation

Before implementing surrogate architectures, the first phase focused on building a reliable numerical simulation pipeline validated end-to-end against established benchmarks:

- **Single-degree-of-freedom validation (EXP1):** Validated elastoplastic dynamics and Bouc-Wen hysteretic loops against analytical solutions and verified OpenSeesPy benchmark models.
- **Continuous hysteretic energy tracking (EXP2):** Tracked continuous hysteretic energy dissipation $E_h(t) = \int \mathbf{f}_{int} \cdot d\mathbf{u}$ as a physically grounded validation metric reflecting cumulative structural damage rather than simple peak displacements.
- **Zero-leakage data partitioning (EXP3):** Discarded random train/test splits, which allow models to interpolate across similar structural geometries. Instead, partitioned the dataset simultaneously by structural archetype and ground motion identity.

This groundwork proved essential: every unexpected result encountered downstream traced back to representational or physical characteristics rather than arbitrary training hyperparameters.

2. Problem formulation

A multi-story shear building subjected to horizontal ground acceleration $a_g(t)$ is governed by the second-order matrix equation of motion:

$$\mathbf{M} \ddot{\mathbf{u}}(t) + \mathbf{C} \dot{\mathbf{u}}(t) + \mathbf{f}_{int}(\mathbf{u}(t), \dot{\mathbf{u}}(t)) = -\mathbf{M} \mathbf{1} a_g(t) \quad (1)$$

where $\mathbf{M}, \mathbf{C} \in \mathbb{R}^{N \times N}$ are the mass and Rayleigh damping matrices, $\mathbf{f}_{int}$ is the nonlinear restoring-force operator, $\mathbf{u}(t) \in \mathbb{R}^N$ represents relative floor displacements, and $\mathbf{1} = [1, \dots, 1]^T$. The surrogate objective is to learn an operator mapping the input ground acceleration and structural parameters to the full multi-story response trajectory. Operator learning is uniquely suited here because it enables resolution-invariant evaluation across arbitrary continuous time steps without retraining.

3. Buildings, ground motions, and partitioning scheme

Structures are idealized as lumped-mass shear frames: story mass 30,000 kg, inter-story stiffness 15–60 MN/m, story height 3.5 m, 5% Rayleigh damping tuned to the first two modal frequencies, and bilinear elastoplastic restoring force with 5% kinematic hardening. Six archetypes span fundamental natural periods from 0.35 s to 1.20 s:

Archetype	Stories	T_1 (s)	Structural Role in Experimental Pipeline
3S_T035	3	0.35	Training / In-distribution (stiff 3-story)
3S_T060	3	0.60	Training / In-distribution (medium 3-story)
3S_T090	3	0.90	Training / In-distribution (flexible 3-story)
5S_T055	5	0.55	Training / In-distribution (stiff 5-story)
5S_T085	5	0.85	Training / In-distribution — upper boundary of training period envelope
5S_T120	5	1.20	Held-out OOD-B — unseen flexible archetype (primary extrapolation test)

Earthquake acceleration records were selected from the PEER NGA-West2 database. Partitioning along independent axes of structural archetype and earthquake event yielded four distinct evaluation partitions:

Partition	Structural Group	Earthquake Records	n	Scientific Evaluation Target
In-distribution (ID)	Known archetypes ($T_1 \leq 0.85$ s)	Training records (RSN0001–08)	120	Ordinary within-distribution accuracy
OOD-A	Known archetypes ($T_1 \leq 0.85$ s)	Held-out records (RSN0011–12)	300	Unseen earthquake record generalization
OOD-B	Unseen archetype (5S_T120, $T_1 = 1.20$ s)	Training records (RSN0001–08)	240	Extrapolation across structural period gap (+0.35 s)
OOD-C	Unseen archetype (5S_T120, $T_1 = 1.20$ s)	Held-out records (RSN0011–12)	60	Combined structural + seismic extrapolation

4. First attempt: fixed-grid FNO2D and its failure (EXP4)

The baseline model followed the standard 2D Fourier Neural Operator architecture (Li et al., 2021) defined on a 5×1024 spatiotemporal tensor. Shorter 3-story buildings were zero-padded to fit the 5-story grid. While 5-story buildings achieved a respectable 19.29% median Relative L_2 error, 3-story buildings failed catastrophically: **99.60% median error**.

The failure is rooted in harmonic analysis. A discrete 2D spatial Fourier transform enforces periodic boundary conditions. Padding real floors with zeros creates a sharp step discontinuity at the boundary. The global Fourier spectral kernels interpret this step as high-frequency energy, producing Gibbs ringing that contaminates the real floors below. This was not an optimization failure: tuning hyperparameters or increasing depth cannot resolve an architectural violation of structural topology.

Representation Strategy	3-Story Median Relative L_2	Parameter Count	Diagnostic Outcome
Fixed-grid FNO2D (EXP4, zero-padded)	99.60%	4,735,187	Discontinuity causes severe Gibbs ringing across lattice
Topology-native GNO (EXP5, no padding)	22.09%	674,115	Graph message-passing respects true physical floor boundaries

5. A topology-native fix: Spatiotemporal Graph Neural Operators

To eliminate artificial boundary steps, EXP5 discarded Cartesian grids in favor of topology-native graphs: each building is represented as $G = (V, E)$ where nodes correspond to floor slabs ($|V| = N_{\text{stories}}$) and edges connect physically adjacent floors, weighted by inter-story column stiffness. Each spatiotemporal block integrates spatial graph message-passing:

$$\mathbf{m}_v^{(l+1)}(t) = \varphi \left(\mathbf{h}_v^{(l)}(t), \sum_{u \in N(v)} \Psi(\mathbf{h}_u^{(l)}(t), \mathbf{e}_{uv}) \right) \quad (2)$$

with a node-wise 1D temporal Fourier spectral convolution to capture dynamic wave propagation:

$$\mathbf{h}_v^{(l+1)}(t) = \varphi \left(\mathbf{W} \mathbf{h}_v^{(l+1)}(t) + \mathcal{F}^{-1} \left[\mathbf{R}_f(k) \cdot \mathcal{F}(\mathbf{h}_v^{(l+1)})(k) \right](t) \right) \quad (3)$$

This architecture dropped 3-story median error from 99.60% to **22.09%** while reducing parameter count by 85.8% (from 4.74M to 674K parameters).

6. What graph topology alone cannot resolve: The phase drift dilemma

While the graph operator eliminated padding artifacts, a subtle out-of-distribution failure surfaced when evaluating the unconditioned model on the held-out flexible archetype 5S_T120 ($T_1 = 1.20$ s). Despite predicting peak displacement within **35.21%**, the full-trajectory Relative L_2 error was **115.70%**.

Forensic FFT spectral analysis of predicted trajectories explained the discrepancy: the unconditioned model oscillated at approximately 1.12 Hz (the edge of its training envelope, corresponding to $T_1 \approx 0.89$ s) rather than tracking the true structural resonance at 0.83 Hz.

When two sinusoidal signals share identical amplitudes but differ in frequency by Δf , their relative phase drifts continuously over time. Once phase opposition is reached (anti-phase), the pointwise Euclidean distance between them yields a mathematical relative error of $\sqrt{2} \approx 141\%$. Over a 20.48-second earthquake record, a frequency offset of 0.29 Hz causes several full cycles of phase drift. The model correctly identifies the response energy envelope, but its unconditioned Fourier kernels lock onto training-set frequencies.

7. Conditioning on pre-earthquake modal invariants via FiLM

Before an earthquake occurs, a structure's mass matrix $\mathbf{M}$ and elastic stiffness matrix $\mathbf{K}$ are fully determined from engineering drawings. By solving the generalized undamped eigenvalue problem:

$$\mathbf{K} \boldsymbol{\varphi}_i = \omega_i^2 \mathbf{M} \boldsymbol{\varphi}_i, \quad T_i = 2\pi / \omega_i \quad (4)$$

we extract the building's modal invariants (T_1 – T_3 , ω_1 – ω_3). Because these depend solely on pre-earthquake physical parameters, injecting them into the surrogate leaks zero information regarding future ground motions or response trajectories. Using Feature-wise Linear Modulation (FiLM; Perez et al., 2018):

$$[\boldsymbol{\gamma}, \boldsymbol{\beta}] = \text{MLP}(\mathbf{c}), \quad \mathbf{h}^{(l+1)} = (1 + \boldsymbol{\gamma}) \odot \mathbf{h}^{(l)} + \boldsymbol{\beta} \quad (5)$$

we modulate both the spatial graph layers and temporal Fourier spectral channels across all four operator blocks, dynamically steering spectral weights to the building's resonant frequencies.

Architecture Variant	Params	ID Peak	OOD-A Peak	OOD-B Peak	OOD-C Peak
EXP5 Baseline GNO (unconditioned)	674,115	12.70%	15.86%	35.21%	38.66%
EXP6-B T_1 -conditioned GNO	725,059	2.09%	8.81%	13.47%	14.36%
EXP6-C Multi-modal GNO (T_{1-3} , ω_{1-3})	727,619	8.87%	7.63%	13.06%	17.01%
EXP6-D Shuffled T_1 (falsification ablation)	725,059	2.65%	8.94%	24.33%	36.16%

Peak displacement error across evaluation partitions. Modal conditioning provides significant accuracy improvements on held-out structural archetypes.

Falsification ablation analysis: To confirm that FiLM modulation leveraged genuine physical relationships rather than simply providing auxiliary capacity, EXP6-D permuted modal vectors randomly across batch samples. Under shuffled conditioning, OOD-B peak error deteriorated from 13.47% to **24.33%**, and OOD-C error worsened from 14.36% to **36.16%**. This degradation confirms that the surrogate actively exploits modal correspondence to guide spectral predictions.

8. Computational performance and inference benchmarks

Inference speed and throughput were measured on Apple Silicon GPU (Metal Performance Shaders, MPS) with process-wide synchronization locks to prevent concurrent thread interference:

Model / Execution Engine	Batch Size	Median Latency	Throughput	Speedup vs. OpenSeesPy
OpenSeesPy 5-story NLTHA (reference)	1	54.68 ms	18.3 sim/s	1.00× (baseline)
Fixed-grid FNO2D (EXP4, batch 1)	1	6.60 ms	151.5 sim/s	8.28×
Topology GNO (EXP5, batch 1)	1	16.25 ms	61.5 sim/s	3.36×
T_1 -conditioned GNO (EXP6-B, batch 1)	1	21.45 ms	46.6 sim/s	2.55×
T_1 -conditioned GNO (EXP6-B, batch 32)	32	30.19 ms	1,060 sim/s	~58× (throughput vs single)

Measured over 50 evaluation iterations per configuration. The single-sample latency provides interactive real-time performance, while the batched throughput enables regional-scale portfolio screening.

9. Architecture summary

Architectural Component	Implementation Details and Physics Grounding
Pre-earthquake Modal Engine	Eigenvalue decomposition of $[\mathbf{K}, \mathbf{M}] \rightarrow$ natural periods T_{1-3} and frequencies ω_{1-3} . Zero future response data required.
Topology-Native Graph	$G = (V, E)$: $ V = N_{\text{stories}}$ floor slabs, edges = columns carrying inter-story stiffness. No zero-padding.
Spatiotemporal Blocks ($\times 4$)	Coupled spatial graph message-passing and 1D temporal Fourier spectral convolutions.
FiLM Conditioning Modules	MLP projects modal invariants $\mathbf{c}$ to affine scale ($\boldsymbol{\gamma}$) and shift ($\boldsymbol{\beta}$) parameters modulating all layers.
Multi-Task Output Head	Projects latent states to floor displacements $u_i(t)$, inter-story drift ratios, and peak structural demands.

10. Where I'd take this next

- **Causal state-space models (S4, Mamba):** Replacing static 1D global Fourier layers with causal state-space layers is the single most critical next step. Global Fourier layers cannot adapt their frequency basis dynamically during strong plastic yielding; causal state-space models process time sequentially and can track stiffness degradation.
- **Continuous-time Neural ODE integrators:** Coupling spatial graph message-passing with a neural ODE integrator to model instantaneous elastoplastic constitutive relationships directly.
- **Dynamic modal tracking:** Updating modal conditioning vectors dynamically as periods elongate during severe earthquake shaking.
- **Extension to 3D asymmetric buildings:** Expanding the graph representation to full 3D frames with bidirectional ground shaking and torsional coupling.
- **Tighter falsification controls:** Implementing dimensionally matched Gaussian noise controls to establish rigorous bounds on conditioning sensitivity.

11. Reproducibility & open-source verification

Data partitions were programmatically verified to ensure zero leakage across structural and ground motion identities. Model checkpoints are frozen and hash-verified against raw evaluation logs. The full test suite (305 unit tests) passes in full. All numerical results cited throughout this document are computed live from canonical CSV logs under `results/experiments/exp6/evaluation/` and documented in `results/canonical/canonical_results.json`.

References

- Li, Z., Kovachki, N., Liu, B., Bhattacharya, K., Stuart, A., & Anandkumar, A. (2021). Fourier Neural Operator for Parametric Partial Differential Equations. *International Conference on Learning Representations (ICLR)*.
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